Code

assignment4.py > ...

```python
def fibonacci(n, memo):

    if n == 0:
        return 0

    if n == 1:
        return 1

    if n in memo:
        return memo[n]

    memo[n] = fibonacci(n - 1, memo) + fibonacci(n - 2, memo)
    return memo[n]

def assignment4():
    n = int(input("Enter n: "))
    memo = {}
    print("fibonacci number:", fibonacci(n, memo))
    print(memo)
assignment4()
```

Output

Problems
Output
Debug Console
Terminal
Ports
Python
PS C:\Users\PRANAV\Desktop\pranav python> & C:\Users\PRANAV\AppData\Roaming\uv\python\cpython-3.14.3-windows-x86_64-none\python.exe "c:/Users/PRANAV/Desktop/pranav python/assignment4.py"
Enter n: 6
fibonacci number: 8
{2: 1, 3: 2, 4: 3, 5: 5, 6: 8}
PS C:\Users\PRANAV\Desktop\pranav python>